B. Recursive Queries

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, data structures, dfs and similar | https://codeforces.com/problemset/problem/932/B

Let us define two functions *f* and *g* on positive integer numbers.

$$f(n) = \text{product of non-zero digits of } n$$

$$g(n) = \begin{cases} n & \text{if } n < 10 \\ g(f(n)) & \text{otherwise} \end{cases}$$

You need to process *Q* queries. In each query, you will be given three integers *l*, *r* and *k*. You need to print the number of integers *x* between *l* and *r* inclusive, such that *g*(*x*) = *k*.

Input

The first line of the input contains an integer *Q* ($1 \le Q \le 2 \times 10^5$) representing the number of queries.

Q lines follow, each of which contains 3 integers *l*, *r* and *k* ($1 \le l \le r \le 10^6$, $1 \le k \le 9$).

Output

For each query, print a single line containing the answer for that query.

Examples

Input

```
4
22 73 9
45 64 6
47 55 7
2 62 4
```

Output

```
1
4
0
8
```

Input

```
4
82 94 6
56 67 4
28 59 9
39 74 4
```

Output

```
3
1
1
5
```

Note

In the first example:

- $g(33) = 9$ as $g(33) = g(3 \times 3) = g(9) = 9$
- $g(47) = g(48) = g(60) = g(61) = 6$
- There are no such integers between 47 and 55.
- $g(4) = g(14) = g(22) = g(27) = g(39) = g(40) = g(41) = g(58) = 4$